22BCON1068
Ankit Kumar

Assignment - 05

Date : 30/11/24 41

Subject : Computer Graphics

• Section - A

1. What is vanishing point ?

Ans A vanishing point is a point in a perspective drawing where parallel lines appear to coverage. it is used to represent depth and distance in 3D space on a two - dimentional plane.

2. Mention the two approaches for hidden surface elimination or visible surface detection.

Ans Hidden surface elimination:-
(i) Object - space Methods :- These methods work by comparing objects or parts of objects to determine which are visible. Ex:- Back - Face Culling.
ii) Image - space Methods :- These methods determine the visibility of objects for every pixel on the screen. Exa :- Z - Buffer Algo.

3. What is perspective forshortening ?

Ans Perspective forshortening is the effect of objects appearing smaller as their distance from the viewer increases. This Phenomenon is based on the principles of perspective projection, where Parallel lines converge toward the vanishing point, and the scale of objects diminishers with increasing distance.

4. Give one advantages and one dis-advanges of parallel projection.

Ans Parallel projection :
• Advantage :- Accurate measurements
• Disadvantage :- No depth perception.

5. Define orthographic projection.

Ans Orthographic projection is a type of parallel projection where the projection lines are perpendicular to the projection plane. it is used to represent 3D objects in two dimensions, maintaining accurate dimensions and shapes without perspective distortion.

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• Section - B (3×7=21)

1. What is projection ? Explain in detail.

Ans Projection is the method of representing a 3D object on a 2D surface, such as paper or a screen. It is essential in computer graphics and drawings to visualize objects.

• Types of Projections :

(1) Parallel projection - in parallel projection, the projection lines are parallel to each other and perpendicular to the projection plane. This method is primarily used for technical drawing because it maintains the true dimensions and proportions of the objects.

Types of parallel projection:-

(i) Orthographic Projection:- The projection lines are perpendicular to the projection plane.

example view:- top view, front view, side view

ii) Oblique projection :- The projection lines are inclined at an angle to the projection plane.

example :- Cavalier & Cabinet projections.

iii) Isometric projection :- A specific type of axonometric projection where the angles b/w the axes are equal (120°). It is commonly used to represent 3D objects in Engg. drawing.

(2) Perspective Projection:- In Perspective Projection, the projection lines converge at a single point called the vanishing point, which mimics how the human eye perceives objectives. Objects farther away appear smaller, creating a sense of depth.

Types of Perspective Projection:-

• One-point Perspective :- A single vanishing point.
• Two-point perspective:- Two vanishing points.
• Three-point perspective - Three vanishing points.

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- Advantage:

(i) Parallel projection
Maintain accurate portions
simple & easy to interpret.

(ii) Perspective Projection
- produce realistic images
- provide a sence of depth & spatial Relationship.

- Application of projections:-

→ Engineering and architecture - creating Technical drawing
→ Computer graphics - Rendering 3D Senes on 2D screen.
→ Art and Design - Drawing realistic perspetive.
→ Cartography - Mapping 3D geographial data onto flat surfaces.

2. Explain Parallel Projection. Give advantage & disadvantage of Parallel Projection.

Ans Parallel projection is a method where projection lines are Parallel and intersect the projection plane at a fixed angle. it preserves the true dimensions of objects without Perspective distortion.

Types of Parallel projection:-

i) Orthographic projection:- Lines are perpendicular to the Projection plane.
ex:- Top, front, and side views.

ii) Oblique projection:- Lines are inclined to the projection plane.
ex:- Cavalier and Cabinet Projections.

iii) Isometric Projection:- Equal angle (120°) betn. the axes, showing a 3D view.

- Advantages :-

→ Maintains true proportions and measurements.
→ simple and easy for technical drawing.

- Disadvantages:-

→ Lacks depth and realism.
→ Objects appear flat and less natural.

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3. What is perspective projection? What are types of perspective projection?

Ans. Perspective projection is a method of representing 3D objects on a 2D plane where projection lines converge at one or more vanishing points. This creates a realistic sense of depth, as objects appear smaller when farther away.

- Types of Perspective projection.

(i) One-point perspective:
- A single vanishing point
- Common for straight rods or tunnels.

(ii) Two-point Perspective:
- Two vanishing point
- Often used for building and structures.

(iii) Three-point perspective:
- Three vanishing point
- Used for all objects viewed at an angle, like skyscrapers.

- Section-C

1. Explain the Back-face removal algorithm in detail.

Ans The Back-face removal algo., also known as Back-face Culling, is used in 3D computer graphics to identify and discard the faces of an object that are not visible to the viewer. This optimizes rendering by avoiding unnecessary processing of hidden surfaces.

- Purpose:
  - To improve rendering efficiency.
  - Reduces the number of polygons processed for visible surfaces.

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Concept : When viewing a closed 3D object, the faces pointing away from the viewer (back-faces) are not visible. These can be identified using the Normal vector of the face and the viewing direction.

Steps in Back-face Removal :-

1. Define a face's Normal vector:
   - The normal vector (N) of a polygon face is calculated using the cross-product of two edges of the polygon.

$$N = (V_2 - V_1) \times (V_3 - V_1)$$

where V_1, V_2, V_3 are vertices of polygon.

2. Determine the viewing vector:
   - The view vector (V) is the dirn from the viewer's position to the polygon.
3. Calculate the Dot product :-

$$D = N \cdot V$$

4. Visibility Test :
   - if $D > 0$: The face is visible
   - if $D \leq 0$: The face is a back-face and can be culled.

- Advantages :
  - Eliminates unnecessary computation for hidden surfaces.
  - Reduces rendering time and memory usage.
- Disadvantages :-
  - Ineffective for non-closed objects or transparent surfaces.

- Applications :
  - Used in 3D rendering pipelines for real-time graphics (e.g. gaming, simulation)
  - Integrated into API like OpenGL and Direct3D.

7. Explain the Z-Buffer algorithm in detail.

Ans The Z-Buffer algorithm is a widely used method for hidden surface removal in 3D graphics. it ensures that only the visible surfaces of objects are rendered by keeping track of the depth of each pixel.

Concept : The algo uses a z-buffer (depth-buffer), a 2D array that stores the depth (z-coordinate) of the closest object at each pixel position. During rendering the algorithm compares the depth of incoming pixels and updates the buffer accordingly.

Steps in Z-buffer Algorithm :-

i) Initialization :
- Set all z-buffer value to the maximum depth (e.g. infinity)
- Clear the frame buffer (used for color values).

ii) For each polygon :
- Resterize the polygon to determine the pixels it covers.
- for each pixel :
  → Calculate its depth (z) using interpolation.
  → Compare z with the current value in the z-buffer for that pixel.

iii) Depth Comparison :
- if z is smaller (closer to the viewer), update :
  - z-buffer with the new depth value.
  - Frame buffer with the color of the polygon at that pixel.
- If z is larger, discard the pixel.

- Advantages :- Handles complex scenes with overlapping objects.
  → simple to implement & work in real-time rendering.
- Disadvantages :- Required additional memory for z-buffer.
  → Does not handle transparency or anti-aliasing directly.

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Q. Explain the scan-line algorithm in detail.

Ans The scan-line algorithm is a hidden surface removal method that processes one horizontal line (scan line) at a time to determine visible surfaces in a 3D scene.

Concept :- Instead of processing pixel by pixel, the algorithm works on a row of pixels (scan line) to determine which object is visible at each position along the line.

Steps in the Scan-line Algorithm :-

i) initialize buffers:
- use a depth buffer to store the closest depth value for each pixel on the scan line.
- use a color buffer to store the color of the visible surface.

ii) Process Polygon :-
- for each scan line, identify all polygons intersecting it.
- sort these polygons by depth at each pixel along the scan line.

(iii) Depth comparison :-
for each pixel on the scan line:
- compare the depth (z) value of overlapping polygons.

(iv) Repeat for all scan lines :-
- process all scan lines from top to bottom to complete the rendering.

- Advantages :-
↳ Efficient for rendering scene & Handles multiple visible polygons along scan line.
- Disadvantages :-
↳ complex to implement & Require significant memory for depth and color buffers.

- Applications :- ↳ used in raster graphics.
↳ Suitable for scenes with simple and planar surfaces.